

Saira Atta

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LinkedIn: [linkedin.com/in/sairaatta](https://www.linkedin.com/in/sairaatta) — GitHub: github.com/sairaatta

PROFESSIONAL PROFILE

Software Developer and **AI Engineer** with experience in Python, React.js, OpenAI API, LangChain, and LangGraph. Skilled in developing LLM applications, AI agents, Retrieval-Augmented Generation (RAG) systems, and agentic workflows. Experienced in prompt engineering, tool calling, multi-agent systems, retrieval, evaluation, verification, AI guardrails, and structured outputs. Strong interest in Agentic AI, Generative AI, multimodal systems, and scalable production-oriented AI solutions.

EDUCATION

COMSATS University Islamabad

2021 – 2025

Bachelor's in Computer Science (GPA: 3.64)

Pakistan

PROFESSIONAL EXPERIENCE

AI Engineering Intern @ TkXel

July 2026 – Present

Bahawalpur, Pakistan

Onsite

Tools: Python, OpenAI API, AI Agents, LangChain, LangGraph, RAG, FastAPI

- Developed **LLM applications and AI agents** using Python, OpenAI API, prompt engineering, and tool calling.
- Built **Agentic AI workflows** using ReAct, sequential, parallel, routing, orchestration, and evaluator patterns.
- Developed **Retrieval-Augmented Generation (RAG)** pipelines with document processing, chunking, embeddings, vector search, metadata filtering, and grounded generation.
- Implemented **LangChain and LangGraph workflows** with memory, state management, tool integration, conditional routing, and multi-agent coordination.
- Built production-oriented AI applications using **FastAPI, Streamlit, Gradio, LLM evaluation, guardrails, and output validation.**

ReactJS Developer Intern @ Precise Tech

Jan 2025 – May 2025

Canada

Remote

Tools: HTML, CSS, JavaScript, React.js, Bootstrap, Next.js

- Developed responsive frontend interfaces using **React.js, JavaScript, HTML, CSS, and Bootstrap.**
- Contributed to large-scale web applications, including distribution management systems.
- Implemented reusable UI components and responsive layouts for web applications.

Frontend Developer Intern @ DevelopersHub Corporation

Nov 2024 – Jan 2025

Islamabad, Pakistan

Onsite

Tools: JavaScript, React.js, Vite, Tailwind CSS, Redux

- Developed an e-commerce application using **React.js, Vite, Redux, and Tailwind CSS.**
- Implemented responsive interfaces, reusable components, and client-side state management.

Frontend Development Intern @ Social Swirl

Aug 2024 – Oct 2024

Lahore, Pakistan

Remote

Tools: HTML, CSS, JavaScript, React.js, Vite, Tailwind CSS

- Developed responsive website interfaces using **React.js, JavaScript, HTML, CSS, and Tailwind CSS.**
- Implemented responsive layouts and reusable frontend components.

PROFESSIONAL PROJECTS

DocInsight – RAG-Based Document Intelligence System

Python, OpenAI API, LangChain, LangGraph, ChromaDB

- Built a document question-answering system supporting PDF, DOCX, TXT, and Markdown files.
- Implemented **text chunking, embeddings, vector search, metadata filtering, and document retrieval** for context-aware responses.
- Developed evaluation workflows for **relevance, groundedness, retrieval quality, and answer verification.**
- Added document hashing, caching, and deduplication to reduce repeated processing and improve efficiency.

AI Digital Twin of My LinkedIn Profile

Python, FastAPI, OpenAI API, Pydantic, PostgreSQL

- Developed an **AI Digital Twin of my LinkedIn profile** using a LinkedIn PDF as the source of professional information.
- Implemented **PDF processing, structured profile extraction, and context-aware LLM responses.**
- Built **user/session management, conversation history, caching, and PostgreSQL persistence.**

- Added **prompt injection detection, rate limiting, and security logging** for safer AI interactions.

AI Recipe Assistant

Agentic AI, LangGraph, Groq, MCP

- Built a multimodal AI recipe generator using the **LangGraph Orchestrator Pattern** for dynamic workflow coordination.
- Supported **three input modes**: text-based recipe requests, optional food images using **Moondream** image-to-text, and mood-based recipe recommendations enhanced with a **Weather Tool**.
- Integrated **GPT-OSS 20B via Groq API** and **MCP** for parallel recipe research, orchestrating **N AI chefs for N recipes** with sequential execution.
- Implemented **Pydantic structured outputs, input/output token tracking, logging, and Streamlit UI**.

Project Allocation & Evaluation System

React, Vite, Tailwind CSS, Redux, Django, SQLite

- Designed and implemented a full-stack Final Year Projects Allocation and Evaluation website following the Guidelines of NCEAC Rules provided by HEC.
- Automated the manual project allocation process, allowing students to select groupmates, supervisors, and projects based on their interests.
- Ensured compliance with NCEAC rules and university FYP committee rules for project allocation and progress evaluation.
- The system facilitates streamlined, transparent project management for both students and faculty.

Nature True Green Landscaping Website Development

WordPress, Elementor

- Developed a responsive WordPress website for a Dubai-based landscaping and garden maintenance company.
- Designed a modern, brand-aligned UI with optimized CTAs, contact forms, and WhatsApp integration for lead generation.
- Improved website performance, mobile responsiveness, and SEO structure through image and layout optimization.
- Live Website: naturetruegreenlandscaping.com

Personal Portfolio website

React, Vite, TailwindCSS, TypeScript, EmailJS

- Developed a modern, responsive personal portfolio website to showcase my projects and skills.
- Contact Form integration using EmailJS.
- Styled the interface with Tailwind CSS for efficient and consistent design.
- Designed dynamic navigation and project presentation to enhance user experience.
- Live Website: sairaatta.tech

TECHNICAL SKILLS

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|-----------------------|-----------------------|----------------------|
| • Python | • LLM Engineering | • Prompt Engineering |
| • JavaScript | • Generative AI | • Agent Workflows |
| • TypeScript | • RAG | • Embeddings |
| • React.js | • LangGraph | • Vector Databases |
| • FastAPI / REST APIs | • Tool Calling | • Git/GitHub |
| • AI Agents | • Multi-Agent Systems | • Tailwind CSS |

CERTIFICATIONS

Jan 2025	Frontend Development	DevelopersHub Corporation
Oct 2024	Frontend Development	Social Swirl

LANGUAGES

English, Urdu